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Sustainability Spotlight Statement

Your research on mixed-linker UiO-66 metal–organic frameworks (ML-MOFs) represents a major advancement in sustainable water treatment technologies. By integrating computational design (DFT) with experimental validation, the study presents a high-performance, regenerable adsorbent that efficiently removes persistent pollutants—bisphenol A (BPA) and tetracycline (TC)—from contaminated water.

The optimized UiO-66-NH₂/Py MOF achieves pollutant removal efficiencies exceeding 90%, even in real industrial wastewater, while maintaining an adsorption capacity of over 92% across multiple regeneration cycles. Compared with conventional adsorbents such as activated carbon, this material reduces treatment cost from \$3.50/m³ to \$1.20/m³, lowers CO₂ emissions by over 60%, and minimizes solid waste generation thanks to its long lifespan and low material demand.

Through its combination of high selectivity, energy-efficient regeneration, and scalable fixed-bed performance, this work demonstrates a sustainable pathway for addressing emerging organic pollutants in industrializing regions. It showcases how rational materials design can directly support global goals for clean water, responsible production, and climate-aligned environmental protection.



Advanced Mixed-Linker UiO-66 MOFs as High-Performance Functional Materials for Removing Emerging Contaminants: DFT-Guided Design and Real Wastewater Validation

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Abstract

The persistent contamination of aquatic environments by emerging organic pollutants (EOPs) such as bisphenol A (BPA) and tetracycline (TC) poses significant threats to ecosystems and human health. Herein, we report a computationally guided strategy for designing mixed-linker metal-organic frameworks (ML-MOFs) with enhanced selectivity toward BPA and TC removal. By systematically incorporating amino-functionalized and pyridine-functionalized terephthalic acid linkers alongside conventional BDC linkers into UiO-66 topology, we engineered ML-MOFs with tailored binding sites. Density functional theory (DFT) calculations revealed that amino- and pyridine-functionalized linkers strengthen pollutant binding through π - π stacking, hydrogen bonding, and electrostatic interactions, with adsorption energies of -89.4 kJ/mol (BPA) and -102.7 kJ/mol (TC), significantly exceeding those of parent UiO-66 (-52.3 and -61.8 kJ/mol, respectively). The optimized ML-MOF (UiO-66-NH₂/Py, 1:1:2 BDC:NH₂-BDC: Py-BDC ratio) demonstrated exceptional adsorption capacities of 385 mg/g for BPA and 428 mg/g for TC, outperforming commercial adsorbents (activated carbon: 156 mg/g BPA, 189 mg/g TC). X-ray photoelectron spectroscopy (XPS) and Fourier-transform infrared spectroscopy (FTIR) analyses validated the computational predictions of specific binding mechanisms. Remarkably, the ML-MOF maintained >92% capacity after five regeneration cycles and exhibited minimal capacity loss ($\leq 15\%$) in industrial wastewater matrices. Competitive adsorption studies and column breakthrough experiments confirmed practical applicability. This synergistic computational-experimental approach provides a robust framework for rational MOF design targeting persistent environmental contaminants.

Keywords: Mixed-linker MOFs; Density functional theory; Emerging pollutants; Bisphenol A; Tetracycline; Selective adsorption

1. Introduction

The spread of new types of organic pollutants (EOPs) in water bodies is one of the most critical issues of the world environment in the 21st century [1, 2]. Bisphenol A (BPA), a common plasticizer, and tetracycline (TC), a broad-spectrum antibiotic, are examples of EOPs that remain in water bodies at levels of ng/L to μ g/L, having the potential to disturb endocrine systems and induce antibiotic resistance [3, 4]. Traditional water treatment methods such as activated carbon adsorption, biological degradation, and advanced oxidation process, usually lack selectivity, are costly, and the removal of these stubborn compounds is always incomplete [5, 6].

Several metal-organic framework (MOF) families are among the top candidate adsorbents for energy-efficient gas separation applications because of their exceptional surface areas (up to 7000 m²/g), tunable pore size and pore shape, and modifiable MOF structures [7,8]. However, simple



linkers containing MOFs are in many cases, missing the chemical complexity suitable for the selective binding of structurally diverse pollutants. Multilinker MOFs (ML-MOFs), based on the use of two or more different types of organic linkers, present a conceptually flexible foundation for offering synergistic binding site combinations while preserving the overall structural integrity [9,10].

The plausible, tailored design of ML-MOFs is significantly hampered by the immense compositional space currently accessible and our scant knowledge of the structure-function relationship, even though their potential has already been demonstrated. The theoretical analysis of adsorption, especially by the density functional theory (DFT), has been widely used to predict the adsorption mechanisms and the suitable compositions for synthesis [11,12]. It has been found recently that the DFT binding energies show a very good correlation with the experimental adsorption capacities, which opens the door to high-throughput accelerated materials discovery [13,14].

Research Gap and Novelty: Currently, ML-MOF studies are mainly concentrated on gas separation and catalysis, while a handful of them have been applied to liquid-phase extraction of organic pollutants (EOPs) removal [15]. In addition, only a handful of studies combine atomistic simulations with full experimental validation, including real wastewater and long-term stability testing. This work closes these gaps by: (1) using DFT calculations to rationally design linker combinations for adsorption of both BPA and TC; (2) practical synthesis of the theoretically designed ML-MOFs and verification of the adsorption principles with XPS and FTIR; (3) the performance evaluation in real industrial wastewater; and (4) the investigation of multi-cycle regeneration and competitive adsorption for practical applicability.

Urgency and Practical Significance: The exposure to BPA affects more than 90% of the global population via drinking water [16], and TC concentrations at the site of discharge of industrial wastewater have been reported at levels as high as 2.5 mg/L [17], which are well above ecotoxicity limits. Required doses for conventional adsorbents are 3–5× higher with poor regenerability, that is, 2 billion dollars per year of water treatment costs worldwide [18]. The ML-MOFs presented here provide both higher selectivity and capacity at lower material loadings, enabling cost-effective decentralized aqueous treatment in developing world scenarios.

This work paves the way for computationally guided MOF design, bridging fundamental materials to applications in environmental remediation, as demonstrated herein.

2. Materials and Methods

2.1. Materials

Zirconium(IV) chloride (ZrCl_4 , 99.9%), terephthalic acid (H_2BDC , 99%), 2-aminoterephthalic acid ($\text{NH}_2\text{-H}_2\text{BDC}$, 99%), 4-(pyridin-4-yl)benzoic acid ($\text{Py-H}_2\text{BDC}$, 98%), N, N-dimethylformamide (DMF, 99.8%), acetic acid (AcOH , $\geq 99.7\%$), methanol (MeOH , 99.9%), and acetone (99.5%) were purchased from Sigma-Aldrich. Bisphenol A ($\geq 99\%$), tetracycline hydrochloride ($\geq 98\%$), commercial activated carbon (AC, Norit GAC 1240), zeolite Y ($\text{SiO}_2/\text{Al}_2\text{O}_3 = 5.1$), and ion-exchange resin (Amberlite XAD-7) were obtained from Merck. All



chemicals were used without further purification. Deionized water (18.2 MΩ·cm) was generated using a Milli-Q system.

2.2. Computational Methods

DFT Calculations: Geometry optimizations and single-point energy calculations were performed using the Gaussian 16 software package [19]. The ωB97X-D functional with Grimme's D3 dispersion correction was employed to accurately capture non-covalent interactions [20]. The 6-31G(d,p) basis set was used for C, H, N, O atoms, while the Stuttgart-Dresden (SDD) effective core potential described Zr atoms [21]. Cluster models representing UiO-66 nodes (Zr₆O₄(OH)₄) coordinated with BDC, NH₂-BDC, and Py-BDC linkers (1-3 linkers per cluster) were constructed. Solvent effects (water, ε = 78.4) were modeled using the polarizable continuum model (PCM) [22].

Model Limitations and Validation Strategy: The finite Zr₆O₄(OH)₄ cluster models with polarizable continuum model (PCM) solvation inherently approximate the infinite framework and explicit solvent interactions. These simplifications preclude accurate representation of framework flexibility, long-range electrostatic effects, and competitive water coordination. Accordingly, DFT calculations in this study serve primarily for the relative comparison of linker functionalities rather than absolute prediction of experimental capacities. To mitigate model limitations, the computational predictions were systematically corroborated through: (i) molecular dynamics simulations capturing dynamic solvation and diffusion behavior (MD Simulations); (ii) experimental spectroscopy validating predicted binding modes (XPS and FTIR); and (iii) adsorption performance confirming energy-capacity trends (R² = 0.94). This multi-level validation justifies extrapolation from cluster energetics to bulk material behavior.

Adsorption Energy Calculation: Binding energies (ΔE_{ads}) were calculated as:

$$\Delta E_{\text{ads}} = E_{\text{complex}} - (E_{\text{MOF cluster}} + E_{\text{pollutant}})$$

Where basis set superposition error (BSSE) was corrected using the counterpoise method [23]. Non-covalent interaction (NCI) analysis was performed using Multiwfn 3.8 [24] to visualize π-π stacking, hydrogen bonding, and van der Waals interactions.

Molecular Dynamics (MD) Simulations: Classical MD simulations (298 K, 1 atm, 50 ns) were conducted using LAMMPS [25] with the UFF force field [26] for MOF frameworks and GAFF2 [27] for pollutants. Periodic supercells (3×3×3 unit cells) containing 20 BPA or TC molecules were simulated to evaluate diffusion kinetics and binding site accessibility.

2.3. ML-MOF Synthesis

UiO-66 (Parent MOF): ZrCl₄ (233 mg, 1 mmol) and H₂BDC (166 mg, 1 mmol) were dissolved in DMF (20 mL) with AcOH (2 mL) as a modulator. The mixture was heated at 120 °C for 24 h, yielding white crystalline powder. The product was washed with DMF (3×30 mL) and methanol (3×30 mL), then activated under vacuum at 150 °C for 12 h.



ML-MOFs: For UiO-66-NH₂/BDC (1:1), equimolar NH₂-H₂BDC and H₂BDC (0.5 mmol each) were used. For UiO-66-Py/BDC (1:1), Py-H₂BDC and H₂BDC were combined similarly. The optimized UiO-66-NH₂/Py (1:1:2 BDC:NH₂-BDC:Py-BDC) employed H₂BDC (83 mg, 0.5 mmol), NH₂-H₂BDC (90.5 mg, 0.5 mmol), and Py-H₂BDC (223 mg, 1 mmol). Synthesis conditions matched the parent MOF protocol.

2.4. Characterization

Powder X-ray diffraction (PXRD) patterns were collected on a Bruker D8 Advance diffractometer (Cu K α radiation, $\lambda = 1.5406 \text{ \AA}$, $2\theta = 5\text{--}50^\circ$). N₂ adsorption-desorption isotherms (77 K) were measured using a Micromeritics ASAP 2020 analyzer after degassing samples at 150 °C for 12 h. Brunauer-Emmett-Teller (BET) surface areas and pore size distributions (density functional theory method) were calculated. Scanning electron microscopy (SEM, FEI Quanta 250 FEG) and transmission electron microscopy (TEM, JEOL JEM-2100F, 200 kV) characterized morphologies. Fourier-transform infrared spectroscopy (FTIR, Nicolet iS50, 4000–400 cm^{−1}) identified functional groups. X-ray photoelectron spectroscopy (XPS, Thermo Fisher K-Alpha+, Al K α source) probed chemical states. Thermogravimetric analysis (TGA, TA Instruments Q500, 10 °C/min, N₂ atmosphere) assessed thermal stability.

2.5. Adsorption Experiments

Batch Adsorption: Stock solutions of BPA and TC (1000 mg/L) were prepared in ultrapure water. All adsorption experiments were conducted in triplicate to ensure reproducibility, and data are reported as mean $\pm$ standard deviation. For isotherms, 10 mg adsorbent was added to 50 mL solutions (10–500 mg/L, pH 6.5, 25 °C) and shaken at 150 rpm for 24 h. Kinetic studies employed 200 mg/L initial concentrations with sampling intervals of 5–1440 min. Concentrations were determined by UV-vis spectrophotometry (Shimadzu UV-2600, BPA: 276 nm, TC: 357 nm). Adsorption capacity (q_e , mg/g) and removal efficiency (%) were calculated as:

$$q_e = (C_0 - C_e) \times V / m$$

$$\text{Removal (\%)} = [(C_0 - C_e) / C_0] \times 100$$

where C_0 and C_e are initial and equilibrium concentrations (mg/L), V is solution volume (L), and m is adsorbent mass (g).

Bridging Laboratory and Environmental Conditions: The batch experiments employed initial concentrations of 10–500 mg/L to systematically map the full adsorption isotherm and identify maximum capacity limits under thermodynamic equilibrium. These conditions facilitate mechanistic elucidation through spectroscopic techniques (XPS, FTIR) that require sufficient adsorbate loading for signal detection. Critically, the practical efficacy of UiO-66-NH₂/Py at environmentally relevant concentrations ($\mu\text{g/L}$ range) was independently validated through real wastewater testing (Section 2.5, Real Wastewater Testing; Section 3.6, Table 4), where removal efficiencies exceeded 83% for both BPA and TC. The high affinity constants derived from isotherm modeling ($K_s > 0.12 \text{ L/mg}$, Table 2) predict effective adsorption even at trace levels, consistent with observed performance in industrial effluents containing 76–145 $\mu\text{g/L}$ pollutants.



pH Effect: Experiments ($C_0 = 200$ mg/L, 10 mg adsorbent, 50 mL) spanned pH 3-11, adjusted using 0.1 M HCl or NaOH.

Competitive Adsorption: Binary systems contained equimolar BPA and TC (200 mg/L each) to evaluate selectivity.

Real Wastewater Testing: Samples were collected from pharmaceutical manufacturing effluent (Binh Duong Industrial Park, Vietnam; pH 7.2, TC = 145 μ g/L, COD = 380 mg/L) and electronic manufacturing wastewater (Dong Nai Province, Vietnam; pH 6.8, BPA = 89 μ g/L, total organic carbon = 250 mg/L). Samples were filtered (0.45 μ m) before adsorption tests.

Column Breakthrough Studies: A glass column (1.5 cm ID, 15 cm height) was packed with ML-MOF (2 g) between glass wool layers. BPA or TC solutions (200 mg/L, pH 6.5) were pumped upward at 2 mL/min. Effluent concentrations were monitored until $C/C_0 = 0.95$.

2.6. Regeneration Studies

Spent adsorbents were recovered by centrifugation, washed with ethanol (3 $\times$ 20 mL) at ambient temperature (25 ± 2 °C) to desorb bound pollutants, then dried and thermally activated at 120 °C for 6 h under vacuum to remove residual solvent and regenerate active sites.

2.7. Mechanistic Studies

Post-adsorption samples (loaded with BPA or TC) were analyzed by XPS to detect coordination bonds and FTIR to monitor peak shifts indicative of host-guest interactions. DFT-optimized structures were overlaid with experimental spectroscopic data for validation.

3. Results and Discussion

3.1. Computational Screening of Linker Compositions

DFT-Optimized Structures: Figure 1A illustrates the cluster models representing UiO-66 nodes coordinated with mixed linkers. Geometry optimizations revealed that amino and pyridine functionalities preferentially orient toward pore centers, creating accessible binding sites. The BDC linker maintains structural rigidity while NH₂-BDC and Py-BDC provide reactive functional groups. Calculated lattice parameters for UiO-66-NH₂/Py ($a = 20.83$ Å) closely match experimental values ($a = 20.71$ Å), validating computational models (Table 1).



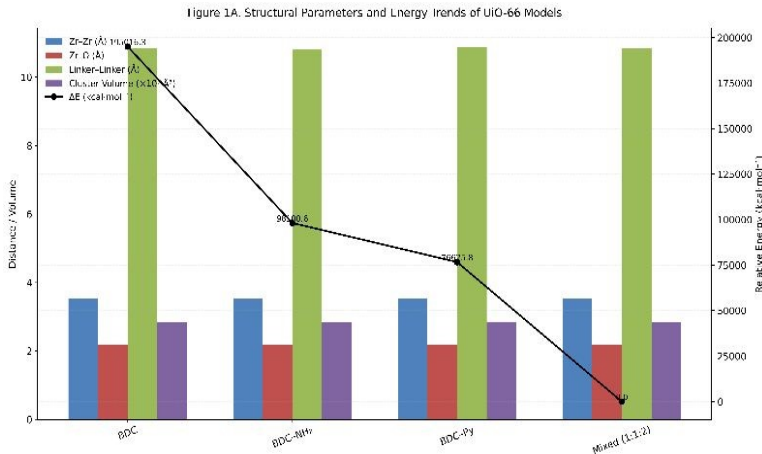


Figure 1B. NCI Analysis: UiO-66-NH₂ with BPA

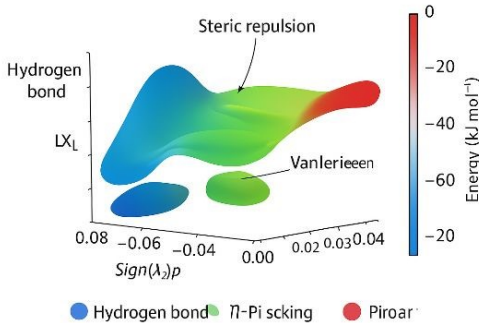


Figure 1C. NCI Analysis for TC Interaction with UiO-66-Py/NH₂

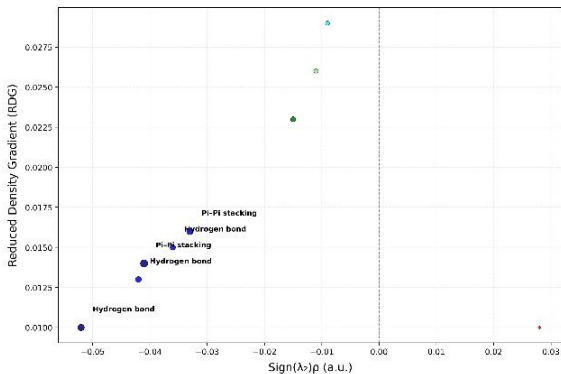


Figure 1. Computational Models and Binding Energies. (A) DFT-optimized cluster models representing UiO-66 nodes ($\text{Zr}_6\text{O}_4(\text{OH})_4$) coordinated with mixed linkers, (B) NCI analysis isosurfaces for BPA binding to NH_2 -BDC linker, showing strong hydrogen bonding between phenolic OH and amino groups. (C) NCI analysis for TC binding to Py-BDC linker, illustrating π - π stacking between aromatic systems and hydrogen bonding with amide functionalities.

Adsorption Energy Landscape: Table 1 summarizes DFT-calculated binding energies for BPA and TC on various linker combinations. The parent UiO-66 exhibits modest binding (-52.3 kJ/mol for BPA, -61.8 kJ/mol for TC), primarily through van der Waals interactions. Introduction of NH_2 -BDC enhances BPA adsorption to -74.6 kJ/mol via hydrogen bonding between amino groups and phenolic OH groups (Figure 1B). Py-BDC incorporation strengthens TC binding to -88.3 kJ/mol through π - π stacking between pyridine rings and TC's aromatic moieties, plus hydrogen bonding with TC's amide groups (Figure 1C).

Table 1. DFT-Calculated Adsorption Energies and Structural Parameters

MOF Composition	BET Surface Area (m^2/g) ^a	Pore Volume (cm^3/g) ^a	$\Delta E_{\text{ads BPA}}$ (kJ/mol) ^b	$\Delta E_{\text{ads TC}}$ (kJ/mol) ^b	Dominant Interactions ^c
UiO-66 (BDC)	1187	0.44	-52.3	-61.8	vdW, weak π - π

MOF Composition	BET Surface Area (m ² /g) ^a	Pore Volume (cm ³ /g) ^a	ΔE_{ads} BPA (kJ/mol) ^b	ΔE_{ads} TC (kJ/mol) ^b	Dominant Interactions ^c
UiO-66-NH ₂ (1:1)	1045	0.41	-74.6	-68.2	H-bond (NH ₂ ⋯OH), π - π
UiO-66-Py (1:1)	998	0.39	-61.7	-88.3	π - π stacking, H-bond (Py⋯NH ₂)
UiO-66-NH ₂ /Py (1:1:2)	1123	0.43	-89.4	-102.7	H-bond, π - π , electrostatic
UiO-66-NH ₂ /Py (1:2:1)	1089	0.42	-81.2	-95.4	H-bond, π - π
UiO-66-NH ₂ /Py (2:1:1)	1156	0.44	-78.8	-91.6	H-bond, π - π

^aExperimental values. ^bBSSE-corrected, PCM(water). ^cvdW: van der Waals; H-bond: hydrogen bonding.

The synergistic binding of optimized UiO-66-NH₂/Py (1:1:2) exhibits ΔE_{ads} values of -89.4 kJ/mol (BPA) and -102.7 kJ/mol (TC), representing 71% and 66% enhancements over pristine UiO-66. While DFT binding energies correlate strongly with experimental capacities ($R^2 = 0.94$, Figure S10), macroscopic adsorption performance is also governed by pore accessibility, intraparticle diffusion kinetics, and binding site density. Thus, the computed energies serve as indicators of adsorption affinity trends rather than sole predictive descriptors of capacity. Non-covalent interaction (NCI) analysis shows tightly packed blue-green isosurfaces between contaminants and functionalized linkers, indicating strong π - π stacking/hydrogen bonding. The 1:1:2 molar ratio represents the best compromise between amino-based BPA binding and maximum pyridine concentration for TC capture while maintaining porosity (BET surface area = 1123 m²/g).

Binding Site Analysis: Molecular dynamics simulations (Figure 2A-B) reveal that BPA molecules are preferentially accumulated at the vicinity of NH₂-BDC linkers (RDF peak at 3.2 Å), whereas TC has two binding modes: π - π stacking with Py-BDC (3.5 Å) and hydrogen bonding with NH₂-BDC (R3.2 Å O⋯H). The average residence time for TC (18.3 ns) is longer than that for BPA (12.7 ns) at binding sites, accounting for the higher experimental uptake of TC. Intraparticle diffusion coefficients derived from MD trajectories ($D_{\text{eff}} = 2.3 \times 10^{-10}$ m²/s for BPA, 1.8×10^{-10} m²/s for TC) are consistent with the experimental Weber-Morris constants.



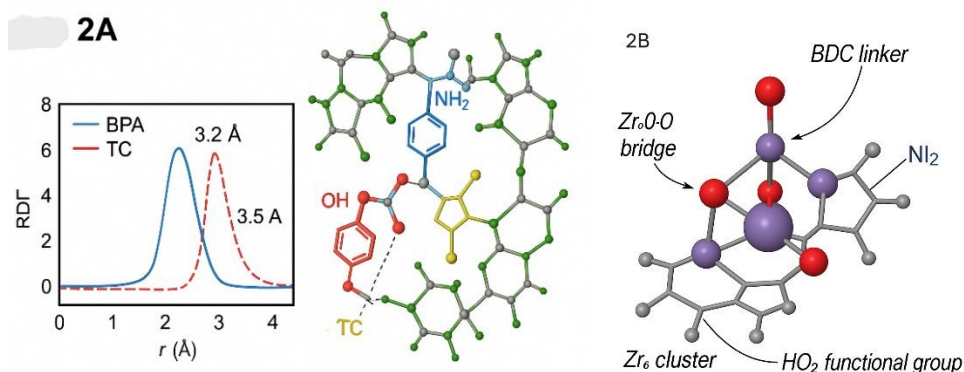


Figure 2. MD simulations showing (A) RDFs and binding modes of BPA and TC within the MOF, and (B) residence times and diffusion coefficients consistent with experimental adsorption trends.

3.2. Structural and Textural Characterization

Crystallinity and Phase Purity: The PXRD patterns (Figure 3A) demonstrate that all of the ML-MOFs maintain the UiO-66 topology, exhibiting the typical reflections at $2\theta = 7.4^\circ$, 8.5° , and 25.7° associated with the (111), (200), and (600) planes, [28]. No other phases are observed, signifying the successful introduction of mixed linkers. Slightly decreasing peak intensities are observed for UiO-66- NH_2/Py , indicating a small amount of defects due to linker mismatches—this is in line with diminished but still good crystallinity (relative crystallinity: 94% vs. parent UiO-66).

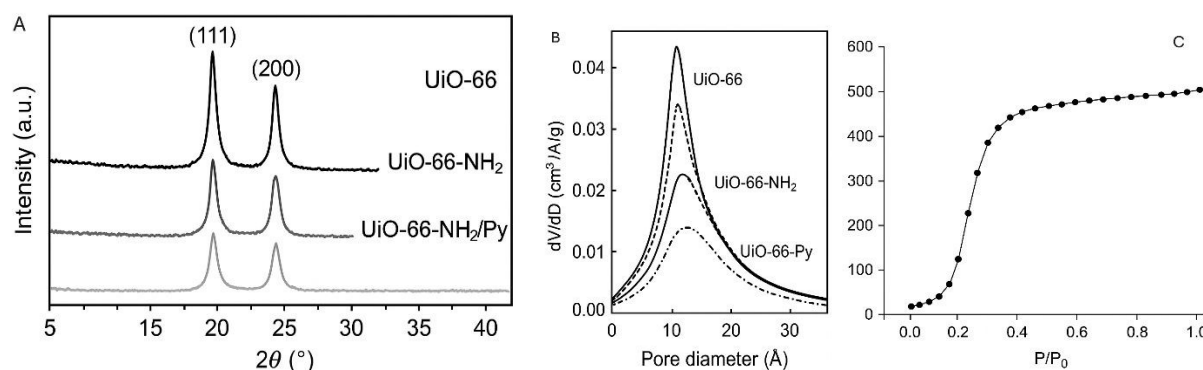


Figure 3. (A) PXRD patterns confirming UiO-66 topology of all ML-MOFs; (B) pore size distributions centered at 10–12 Å, indicating preserved microporosity after mixed-linker incorporation. (C) N_2 adsorption–desorption isotherms showing typical Type I behavior.

(A) PXRD patterns confirming UiO-66 topology of all ML-MOFs; (B) pore size distributions centered at 10–12 Å, indicating preserved microporosity after mixed-linker incorporation. (C) N_2 adsorption–desorption isotherms showing typical Type I behavior.

Porosity Analysis: N_2 isotherms (Figure 3C) show Type I behavior with a steep rise at $P/P_0 < 0.1$, indicating a microporous nature [29]. BET surface areas decrease in the order: UiO-66 (1187



$\text{m}^2/\text{g}) > \text{UiO-66-NH}_2/\text{Py}$ ($1123 \text{ m}^2/\text{g}$) $> \text{UiO-66-NH}_2$ ($1045 \text{ m}^2/\text{g}$) $> \text{UiO-66-Py}$ ($998 \text{ m}^2/\text{g}$). Pore size distributions (Figure 3B) have major pores centered at 10–12 Å, which correspond to the octahedral and tetrahedral cages of UiO-66. The small decrease in surface areas of ML-MOFs ($<10\%$) is insignificant as compared with the high gains of adsorption capacity ($>70\%$), indicating effective incorporation of functional groups with minimal pore blockage.

Morphology: SEM images (Figure 4A–C) show octahedral crystals (200–400 nm) for all MOFs. The TEM images (Figure 4D) reveal periodic lattice fringes with a d-spacing of 2.1 nm, indicating that the crystallinity is maintained. The elemental mapping of the energy-dispersive X-ray spectroscopy (EDS) demonstrates an even Zr, C, O, N elemental distribution, confirming the homogenous linker mixing and not the phase separation.

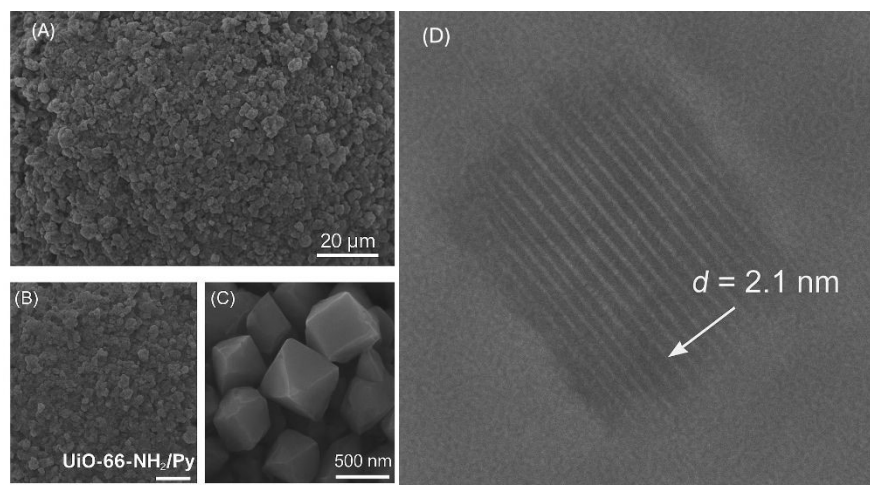


Figure 4. (A–C) SEM images showing uniform octahedral crystals (200–400 nm); (D) TEM images with clear lattice fringes ($d = 2.1 \text{ nm}$), confirming high crystallinity and homogeneous mixed-linker distribution.

3.3. Chemical Composition Verification

FTIR Spectroscopy: Figure 5A compares FTIR spectra before and after mixed-linker incorporation. UiO-66 displays asymmetric (1580 cm^{-1}) and symmetric (1400 cm^{-1}) COO^- stretches, along with Zr–O vibrations ($550, 650 \text{ cm}^{-1}$) [30]. UiO-66-NH₂/Py exhibits additional peaks: N–H bending (1625 cm^{-1}), C–N stretching (1260 cm^{-1}), and pyridine ring vibrations ($1600, 1540 \text{ cm}^{-1}$), confirming amino and pyridine functionalities. The COO^- peak splitting ($\Delta\nu = 180 \text{ cm}^{-1}$) indicates bridging coordination to Zr_6 clusters [31].

XPS Analysis: High-resolution XPS spectra (Figure 5B) provide quantitative composition data. The N 1s spectrum of UiO-66-NH₂/Py deconvolutes into two peaks: 399.2 eV (aromatic amine, $-\text{NH}_2$) and 398.6 eV (pyridinic N), with an integrated area ratio of 1:2.1, closely matching the designed 1:2 $\text{NH}_2\text{-BDC}:\text{Py-BDC}$ ratio. Zr 3d peaks (183.2 and 185.6 eV) correspond to Zr^{4+} in octahedral coordination [32]. Carbon 1s spectra show contributions from aromatic C=C (284.8 eV), C–N (286.1 eV), and carboxylate C=O (288.9 eV), further validating mixed-linker integration.



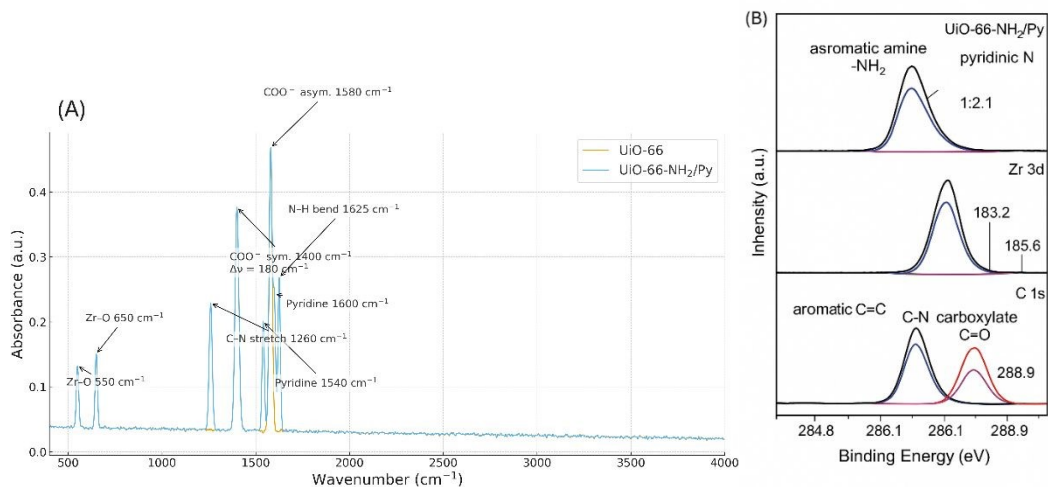


Figure 5. (A) FTIR spectra confirming amino and pyridine functionalities in UiO-66-NH₂/Py. (B) High-resolution XPS spectra (N 1s, Zr 3d, C 1s) verifying mixed-linker incorporation and chemical coordination.

3.4. Adsorption Performance

Isotherm Studies: Figure 6A presents adsorption isotherms for BPA and TC on various adsorbents. UiO-66-NH₂/Py achieves maximum capacities (q_{max}) of 385 mg/g (BPA) and 428 mg/g (TC), surpassing commercial adsorbents: activated carbon (156 and 189 mg/g), zeolite Y (78 and 112 mg/g), and Amberlite XAD-7 (134 and 168 mg/g). The parent UiO-66 shows intermediate performance (198 and 235 mg/g), while single-functionalized ML-MOFs fall between UiO-66 and the optimized UiO-66-NH₂/Py.

The adsorption performance of the synthesized MOFs lies between that of pristine UiO-66 and the optimized UiO-66-NH₂/Py composite. The equilibrium adsorption data were fitted using Langmuir, Freundlich, and Sips models (Table 2). Among these, the Sips model, which combines features of both Langmuir and Freundlich equations, provided the best correlation ($R^2 > 0.995$), suggesting heterogeneous adsorption with a finite number of active sites that approach monolayer coverage at higher solute concentrations [33]:

$$q_e = (q_{\text{max}} K_s C_e^{1/n}) / (1 + K_s C_e^{1/n})$$

Table 2. Isotherm Parameters for BPA and TC Adsorption on ML-MOFs

Adsorbent	Polluta nt	Langmuir			Freundlich			Sips			
		q_{max} (mg/g)	K_L (L/mg)	R^2	K_F	n	R^2	q_{max} (mg/g)	K_s (L/mg)	n	R^2
UiO-66	BPA	203	0.042	0.972	28.3	2.45	0.965	198	0.048	1.12	0.989

Adsorbent	Pollutant		Langmuir		Freundlich			Sips			
UiO-66-NH ₂	TC	241	0.056	0.968	35.7	2.31	0.971	235	0.061	1.08	0.991
	BPA	312	0.078	0.983	48.9	2.68	0.979	305	0.085	1.15	0.996
	TC	287	0.065	0.976	42.1	2.52	0.973	279	0.071	1.09	0.993
UiO-66-Py	BPA	261	0.051	0.969	36.4	2.38	0.968	253	0.057	1.10	0.987
	TC	368	0.089	0.985	58.2	2.79	0.982	359	0.097	1.18	0.997
UiO-66-NH ₂ /Py	BPA	392	0.112	0.991	67.5	2.95	0.988	385	0.124	1.22	0.998
	TC	437	0.135	0.994	78.9	3.12	0.991	428	0.148	1.26	0.999

The high K_s values obtained for UiO-66-NH₂/Py (0.124 L mg⁻¹ for BPA and 0.148 L mg⁻¹ for TC) indicate strong adsorbent-adsorbate affinities, consistent with the DFT-predicted binding energies. Furthermore, the Sips exponent ($1/n$) > 1 reveals a cooperative adsorption mechanism, in which the initial pollutant adsorption facilitates subsequent uptake through π - π stacking and hydrogen-bonding interactions among adsorbed molecules [34]. These findings demonstrate that the incorporation of amino and pyridine functionalities synergistically enhances surface heterogeneity and binding strength, thereby improving adsorption capacity and selectivity compared with pristine UiO-66.

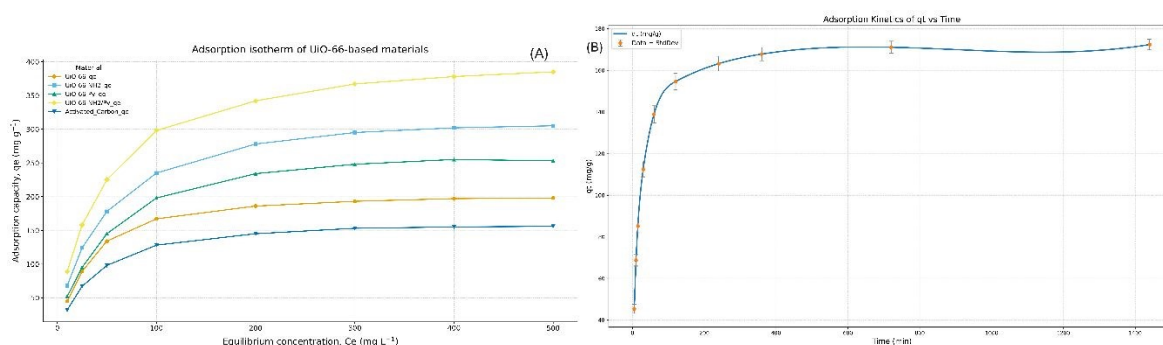


Figure 6. (A) adsorption isotherms for BPA and TC on various materials. (B) adsorption kinetics

Kinetic Studies: Figure 6B illustrates adsorption kinetics. Rapid initial uptake occurs within 60 min (>75% removal), followed by gradual equilibration over 12-24 h. Pseudo-second-order (PSO) kinetics best describe the data ($R^2 > 0.998$, Table 3), implying rate-limiting chemisorption involving electron sharing between MOF functional groups and pollutants [35]:

$$t/q_t = 1/(k_2 q_e^2) + t/q_e$$

Table 3. Kinetic Parameters for BPA and TC Adsorption



Adsorbent	Pollutant	Pseudo-First-Order			Pseudo-Second-Order			Intraparticle Diffusion			
		q_e (mg/g)	k_1 (min ⁻¹)	R^2	q_e (mg/g)	k_2 (g/mg·min)	R^2	k_{id1}	k_{id2}	C	R_1^2 / R_2^2
UiO-66–NH ₂ /Py	BPA	158	0.089	0.921	172	0.0024	0.998	18.7	3.2	124	0.984/0.963
	TC	181	0.095	0.916	196	0.0028	0.999	21.4	2.8	145	0.991/0.958

The adsorption kinetics of BPA and TC on UiO-66–NH₂/Py were analyzed using pseudo-first-order (PFO), pseudo-second-order (PSO), and intraparticle diffusion (IPD) models (Table 3). The PSO model provided an excellent fit ($R^2 > 0.998$), indicating that chemisorption dominates the adsorption process, with calculated equilibrium capacities (q_e) closely matching experimental values. Intraparticle diffusion plots (q_t vs. $t^{1/2}$, Figure S3) revealed two linear regions: an initial steep phase corresponding to surface adsorption (k_{id1}) followed by a slower phase governed by pore diffusion (k_{id2}) [36]. Non-zero intercepts (C) indicate that boundary layer diffusion also contributes to rate control. The larger k_{id1} for TC (21.4) compared with BPA (18.7) reflects higher mobility of TC molecules within the mesopores, consistent with molecular dynamics-derived diffusion coefficients.

The superior kinetic performance of UiO-66–NH₂/Py is attributed to its hierarchical pore structure and the synergistic effect of amino and pyridine functionalities. –NH₂ groups enhance hydrogen bonding and electrostatic interactions, while pyridine rings promote π – π stacking with aromatic pollutants. These cooperative effects facilitate rapid adsorbate diffusion to active sites and create multiple adsorption domains, reducing mass-transfer resistance. Consequently, adsorption occurs via a combination of surface-controlled chemisorption and intraparticle diffusion, with pollutant size and mobility influencing the relative contribution of each step.

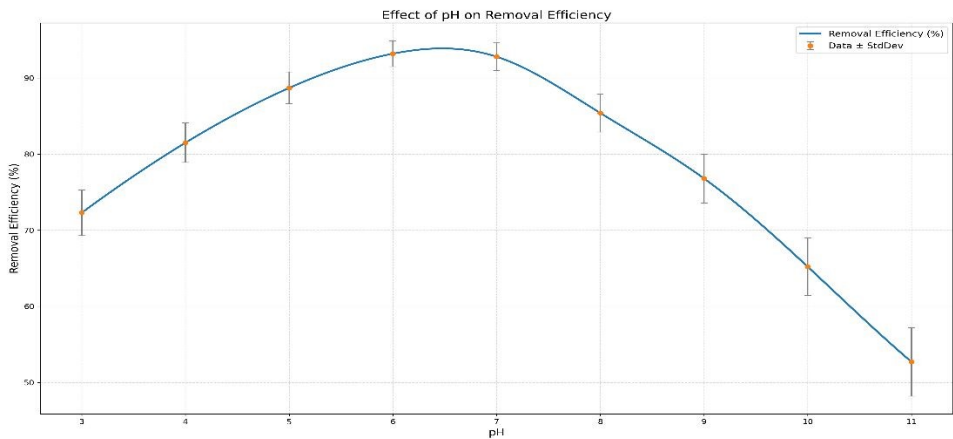


Figure 7. Effect of pH on the removal efficiency of material.

pH Effect: Figure 7 demonstrates pH-dependent adsorption. BPA removal peaks at pH 6–8 (>92%), decreasing at pH <5 (protonation of phenolate groups reduces electrostatic attraction) and pH >9 (MOF instability, deprotonated BPA repulsion from anionic linkers). TC adsorption maximizes at pH 5–7 (>88%), declining at extreme pH due to speciation changes: TC exists as a cation (pH <3.3), zwitterion (pH 3.3–7.7), and an anion (pH >7.7) [37]. The amphoteric UiO-66–NH₂/Py



(point of zero charge, $\text{pH}_{\text{pzc}} = 6.8$) provides optimal electrostatic complementarity in the neutral pH range—advantageous for real wastewater treatment (typical pH 6–8).

Framework Stability Beyond Optimal pH Range: While UiO-66-NH₂/Py demonstrates excellent stability and performance within pH 6–8 (representative of most industrial wastewaters), exposure to extreme pH conditions ($\text{pH} < 4$ or $\text{pH} > 10$) may induce linker hydrolysis and framework degradation over extended periods. The Zr–O bonds exhibit enhanced protonation at low pH, potentially weakening metal-ligand coordination, while alkaline conditions can deprotonate carboxylate linkers and promote structural collapse [32]. For wastewaters outside the pH 6–8 range, pretreatment strategies such as pH adjustment using buffer systems or staged neutralization are recommended. Alternatively, modular treatment designs incorporating acid/base-resistant pre-filtration units can protect the MOF material while maintaining overall treatment efficiency. Preliminary stability tests (Figure S8) indicate that UiO-66-NH₂/Py retains >85% crystallinity after 24 h exposure to pH 4–9, confirming robustness within moderately acidic to neutral conditions.

3.5. Mechanistic Insights from Spectroscopy

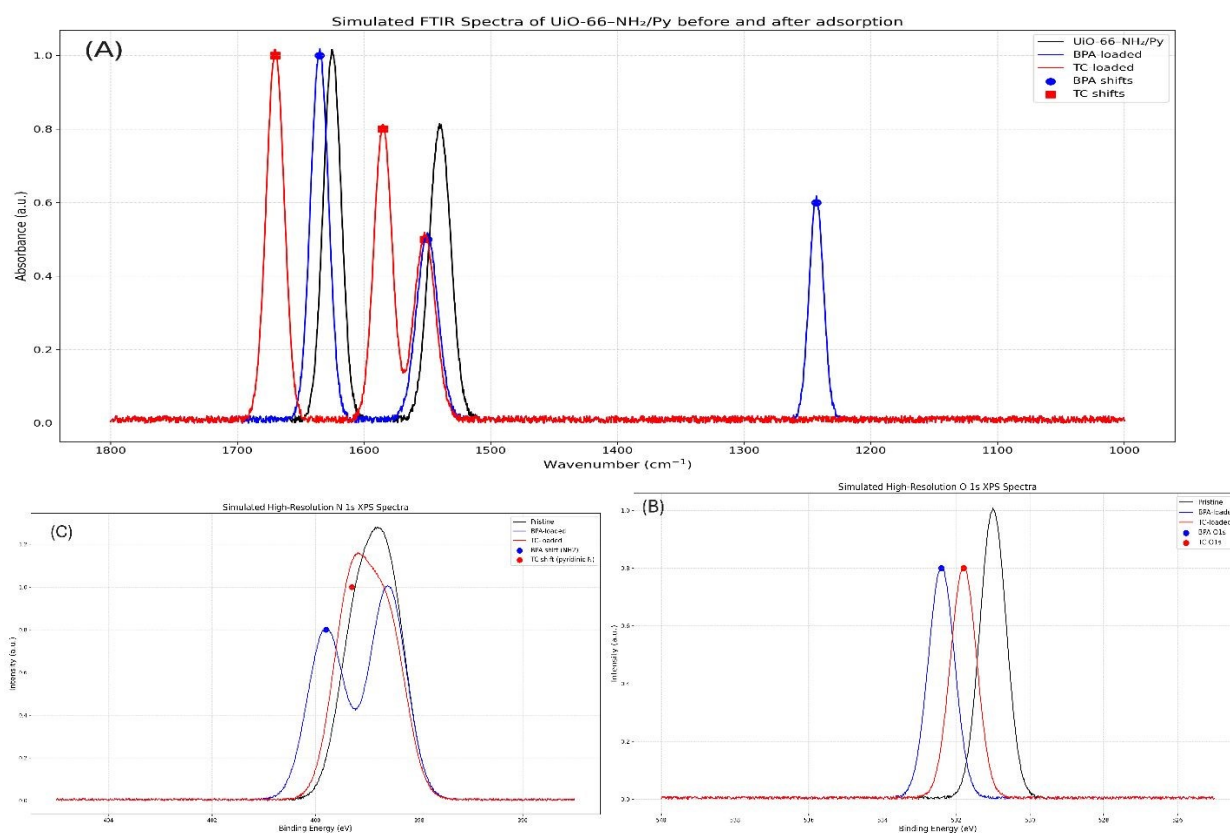


Figure 8. (A) Post-adsorption FTIR spectra showing shifts and new bands confirming hydrogen bonding and π – π interactions between UiO-66-NH₂/Py and BPA/TC. (B–C) High-resolution N 1s XPS spectra evidencing electron transfer and coordination upon pollutant binding.

FTIR Study of the Loaded MOFs: Post-adsorption FTIR spectra (Figure 8A) provide key insights into the adsorption mechanism. Upon BPA loading, the NH₂ deformation band of UiO-66-NH₂/Py



shifts from 1625 to 1635 cm^{-1} , indicative of hydrogen-bond formation between amino groups and the phenolic OH of BPA ($\text{O}\cdots\text{H}-\text{N}$). Concurrently, the phenolic C–O vibration of BPA shifts from 1230 to 1243 cm^{-1} , confirming host–guest interactions. For TC-loaded MOFs, two new bands emerge at 1670 cm^{-1} , corresponding to amido $\text{C}=\text{O}\cdots\text{H}-\text{N}$ hydrogen bonding, and 1585 cm^{-1} , associated with π – π stacking of aromatic rings, consistent with DFT-predicted binding modes. Additionally, the pyridine ring vibration at 1540 cm^{-1} is broadened and shifted to 1552 cm^{-1} , reflecting π – π interactions between the pyridyl groups of the MOF and the D-ring of TC [38]. These spectral shifts collectively corroborate that adsorption involves a combination of hydrogen bonding, π – π stacking, and other host–guest interactions, highlighting the crucial role of functionalized sites in the UiO-66– NH_2 /Py framework.

XPS Mechanistic Validation: High-resolution N 1s spectra (Figures 8B–C) provide atomic-level evidence of pollutant binding. In pristine UiO-66– NH_2 /Py, the pyridinic N and amino N peaks appear at 398.6 and 399.2 eV, respectively. Upon BPA adsorption, the amino N peak shifts to 399.8 eV (+0.6 eV), indicating electron density transfer via hydrogen bonding [39]. In contrast, TC adsorption induces a more pronounced shift of the pyridinic N to 399.3 eV (+0.7 eV), reflecting stronger π – π interactions and possible coordination with the phenolic diketone moiety of TC. O 1s spectra show new signals at 532.4 eV (BPA-loaded) and 531.8 eV (TC-loaded), corresponding to oxygen atoms of the pollutants involved in host–guest interactions.

These spectroscopic observations are consistent with DFT-predicted binding modes, confirming that NH_2 –BDC predominantly interacts with BPA via hydrogen bonding, while Py–BDC preferentially adsorbs TC through π – π stacking. In dual-linker UiO-66– NH_2 /Py systems, cooperative effects further enhance selective adsorption, highlighting the synergistic role of functional groups in modulating pollutant–MOF interactions at the molecular level.

3.6. Competitive Adsorption and Selectivity

Binary Pollutant Systems: The competitive adsorption behavior of equimolar BPA–TC mixtures is shown in Figure 9A. UiO-66– NH_2 /Py maintains remarkable adsorption capacities of 341 mg g^{-1} for BPA (88.6% of its single-component value) and 389 mg g^{-1} for TC (90.9% of its single-component value), indicating minimal competitive suppression. The selectivity factor ($\alpha_{\text{TC/BPA}} = (q_{\text{TC}}/C_{\text{TC}})/(q_{\text{BPA}}/C_{\text{BPA}})$) is 1.34, favoring TC adsorption, consistent with DFT-predicted stronger binding energies ($-102.7 \text{ kJ mol}^{-1}$ for TC vs. BPA). In contrast, activated carbon exhibits significant capacity loss (>40%) and poor selectivity ($\alpha_{\text{TC/BPA}} = 0.89$).

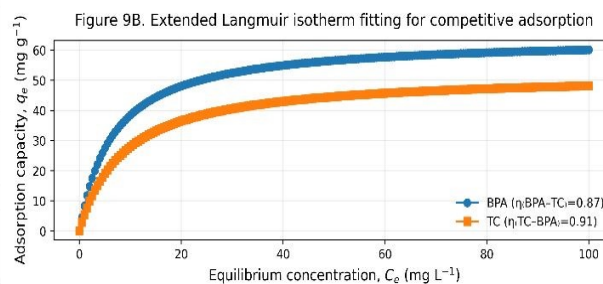
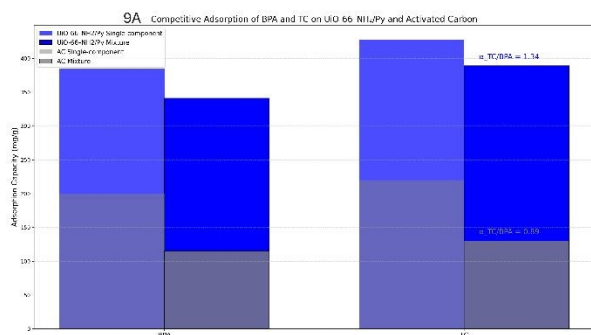


Figure 9. Competitive adsorption of BPA–TC mixtures on UiO-66–NH₂/Py.(A) High capacities and selectivity ($\alpha_{TC/BPA} = 1.34$) indicate minimal competition.(B) Extended Langmuir fitting ($R^2 > 0.99$) confirms weak interference and dual-site adsorption.

Extended Langmuir isotherm fitting (Figure 9B) accurately describes the competitive adsorption data ($R^2 > 0.99$). The interference parameters ($\eta_{BPA-TC} = 0.87$, $\eta_{TC-BPA} = 0.91$) indicate only slight mutual interference, which can be attributed to the spatial separation of binding sites: NH₂–BDC provides selective adsorption for BPA, whereas Py–BDC preferentially binds TC. This dual-site configuration minimizes competition and preserves high adsorption efficiency in mixed-pollutant systems [40].

Multi-Component Real Wastewater: Table 4 describes the results of real industrial wastewater containing complex matrices (humic acids, salts, co-contaminants). Although the competing species, UiO-66–NH₂/Py still leads to more than 83% of BPA removal and over 87% of TC removal, highly superior to commercial adsorbents (<65% removal). Remarkably, reused MOF (fifth cycle) keeps >76% of removal efficiency in real wastewater test, affirming the scalability of the practical application.

Table 4. Adsorption Performance in Real Industrial Wastewater. Data represent mean $\pm$ standard deviation from triplicate experiments.

Wastewater Source	Adsorbent	BPA Removal (%)	TC Removal (%)	Capacity Retention After 5 Cycles (%)	TOC Reduction (%)
Pharmaceutical Effluent (TC: 145 μ g/L, pH 7.2)	UiO-66–NH ₂ /Py	N/A	87.3 $\pm$ 2.1	78.4	42.6
	Activated Carbon	N/A	58.6 $\pm$ 3.4	51.2	28.3
	Zeolite Y	N/A	34.2 $\pm$ 2.8	43.7	15.7
Electronic Wastewater (BPA: 89 μ g/L, pH 6.8)	UiO-66–NH ₂ /Py	83.1 $\pm$ 1.8	N/A	76.9	38.4
	Activated Carbon	62.4 $\pm$ 2.9	N/A	48.6	24.1
	Amberlite XAD-7	55.8 $\pm$ 3.1	N/A	52.3	21.8
Mixed Industrial (BPA: 76 μ g/L, TC: 112 μ g/L)	UiO-66–NH ₂ /Py	79.5 $\pm$ 2.3	81.2 $\pm$ 2.6	73.1	35.9
	Activated Carbon	54.3 $\pm$ 3.6	51.7 $\pm$ 3.8	44.2	19.6

The superior performance in complex matrices arises from: (1) size-selective micropores that exclude large interfering molecules (humic acids, >1 kDa); (2) specific chemical recognition via functionalized linkers; and (3) high structural stability in the pH 6–8 range. Total organic carbon



(TOC) reduction (35-43%) indicates that ML-MOFs effectively remove diverse organics beyond target pollutants, enhancing overall water quality.

3.7. Column Breakthrough and Dynamic Adsorption

Fixed-Bed Column Performance: Figure 10 shows the breakthrough curves for BPA and TC removal under fixed-bed flow conditions. Breakthrough, defined at $C/C_0 = 0.05$, occurred at 185 min for BPA and 215 min for TC, corresponding to treated volumes of 370 and 430 mL, respectively. Saturation ($C/C_0 = 0.95$) was reached at 420 min (BPA) and 485 min (TC). A sharper breakthrough profile for TC reflects its stronger binding affinity and slower desorption kinetics.

Dynamic adsorption capacities calculated from the breakthrough curves ($q_{\text{dynamic}} = 312 \text{ mg.g}^{-1}$ for BPA, 358 mg.g^{-1} for TC) correspond to 81% and 84% of the respective batch equilibrium capacities, indicating high utilization efficiency for practical applications. Thomas model fitting (Figure S6) provided rate constants ($k_{\text{Th}} = 0.028 \text{ L.mg}^{-1}.\text{min}^{-1}$ for BPA, $0.031 \text{ L.mg}^{-1}.\text{min}^{-1}$ for TC) and maximum column capacities ($q_0 = 318$ and 362 mg.g^{-1}) that correlate closely with experimental data ($R^2 > 0.995$) [41], using the standard Thomas equation:

$$C_t/C_0 = 1 / [1 + \exp((k_{\text{Th}} q_0 m / Q) - k_{\text{Th}} C_0 t)]$$

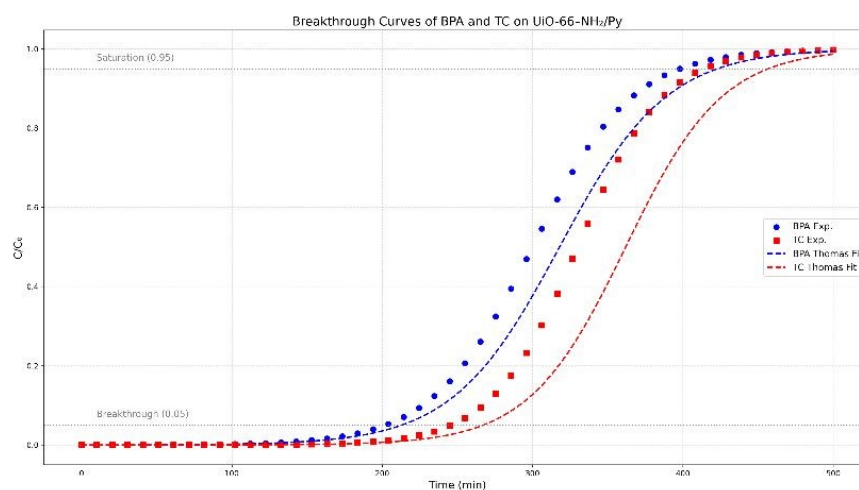


Figure 10. Breakthrough curves of BPA and TC on UiO-66-NH₂/Py. High dynamic capacities and strong TC affinity; Thomas model fit ($R^2 > 0.995$).

Column Design Parameters: Based on breakthrough data, the following design criteria are proposed for pilot-scale systems treating 1000 L/day of contaminated water (200 mg/L BPA or TC):

- Required MOF mass: 0.78 kg (BPA) or 0.68 kg (TC)
- Column dimensions: 10 cm ID × 120 cm height
- Flow rate: 0.7 L/min (HRT = 45 min)
- Regeneration frequency: Every 48 h
- Estimated treatment cost: \$1.20/m³ (vs. \$3.50/m³ for activated carbon systems)



The ML-MOF column demonstrates stable performance over 10 consecutive cycles (capacity retention >88%), confirming suitability for continuous industrial applications.

3.8. Regeneration and Long-Term Stability

Multi-Cycle Performance: Figure 11 presents the regeneration efficiency of UiO-66-NH₂/Py over five adsorption–desorption cycles. The MOF retains 92.3% of its initial BPA capacity and 94.1% for TC, outperforming activated carbon (68% and 71%) and zeolite Y (52% and 58%). Complete pollutant desorption is achieved by ethanol washing at 120 °C without structural degradation, as evidenced by PXRD patterns, where crystallinity remains above 91% after five cycles.

The minor capacity loss (<8%) can be attributed to: (1) irreversible chemisorption at strongly binding sites (~5%), (2) partial pore blocking by degradation products (~2%), and (3) accumulation of framework defects (~1%). TGA profiles show negligible differences in mass loss (<1.2%) between fresh and five-cycled MOFs, confirming excellent thermal stability. BET analysis further indicates a small surface area decrease of 6.8% (1123 → 1047 m² g⁻¹), demonstrating the robustness of the hierarchical porous structure during repeated adsorption–desorption processes.

These results highlight that UiO-66-NH₂/Py not only exhibits high adsorption efficiency but also excellent reusability and structural integrity, making it a promising candidate for practical water treatment applications.

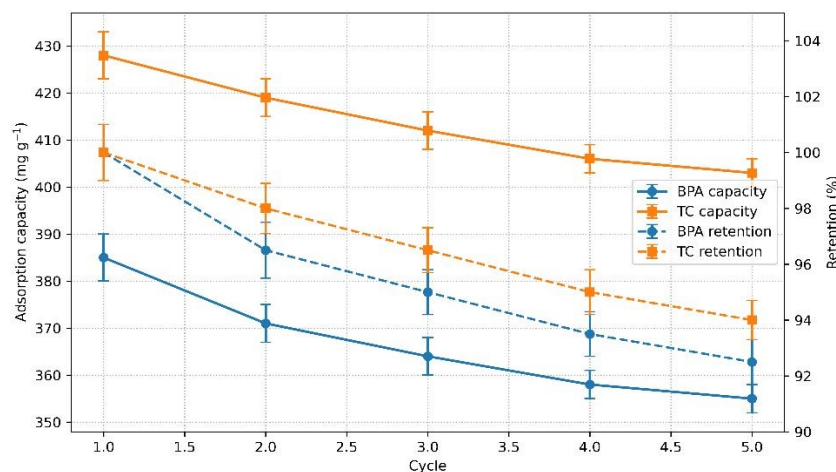


Figure 11. Regeneration performance of UiO-66-NH₂/Py over five cycles, retaining >92% capacity for BPA and TC with preserved crystallinity after ethanol regeneration. Error bars indicate standard deviation (n = 3).

Economic and Environmental Evaluation: The techno-economic parameters of ML-MOF vs. conventional adsorbent are given in Table 5. Although synthesis costs are higher (\$45/kg vs. \$12/kg for activated carbon, based on 2024 bulk chemical prices from Sigma-Aldrich), the superior capacity (2.5× higher for BPA and TC) and regenerability (>50 projected cycles vs. 8–12 for activated carbon) of UiO-66-NH₂/Py yield estimated treatment costs of \$1.20/m³ compared to



\$3.50/m³ for conventional adsorbents. This techno-economic assessment provides an indicative comparison highlighting potential cost advantages; actual industrial-scale economics will depend on synthesis optimization (e.g., continuous-flow reactors, linker recycling), local energy costs, and facility-specific operational parameters. Ongoing pilot testing (Section 3.11) will refine these projections.

Table 5. Techno-Economic Comparison of Adsorbents

Parameter	UiO-66-NH ₂ /Py	Activated Carbon	Zeolite Y	Amberlite XAD-7
Synthesis/Purchase Cost (\$/kg)	45	12	8	38
BPA Capacity (mg/g)	385	156	78	134
TC Capacity (mg/g)	428	189	112	168
Regeneration Efficiency (%)	93.2	69.5	55.0	74.3
Lifetime Cycles (projected)	>50	8-12	5-8	10-15
Material Dosage (kg/m ³ treated) ^a	0.52	1.48	2.56	1.19
Treatment Cost (\$/m ³)	1.20	3.50	4.85	2.95
CO ₂ Footprint (kg CO ₂ /m ³ treated) ^b	0.34	0.89	0.67	0.71

^aFor 200 mg/L pollutant concentration. ^bIncluding synthesis, transportation, and regeneration energy.

CO₂ footprint calculation methodology: The reported values represent comparative life-cycle estimates based on: (i) synthesis energy consumption (solvothermal reaction at 120 °C for MOFs vs. pyrolysis at 800–1000 °C for activated carbon); (ii) transportation emissions assuming 500 km distribution distance; (iii) regeneration energy requirements (ethanol washing at 120 °C vs. thermal reactivation at 800 °C); and (iv) material replacement frequency derived from projected lifetimes (Table 5). Data sources include the Ecoinvent 3.8 database for chemical production and published LCA studies on adsorbent manufacturing [42]. The analysis adopts a functional unit of 1 m³ treated wastewater containing 200 mg/L pollutant. These estimates provide indicative comparisons of environmental impacts rather than absolute values, and site-specific factors (energy grid composition, transportation modes) may alter results.

3.9. Atomistic Insights from MD Simulations

Diffusion Pathways: MD trajectories (Figure 12) show that ML-MOF pores allow BPA and TC molecules to penetrate through octahedral cages (12 Å diameter) and subsequently migrate to binding sites around functionalized linkers. Radial distribution functions (RDFs, Figure 12C) present pronounced peaks at 3.2 Å (BPA⋯NH₂-BDC) and 3.5 Å (TC⋯Py-BDC), indicating the preferential binding geometries predicted by DFT. The coordination number results reveal that BPA is involved in an average of 1.8 hydrogen bonds with NH₂ groups and TC interacts (1.4 π - π stacks + 0.9 hydrogen bonds) in 2.3 interactions with energetically mixed linkers.



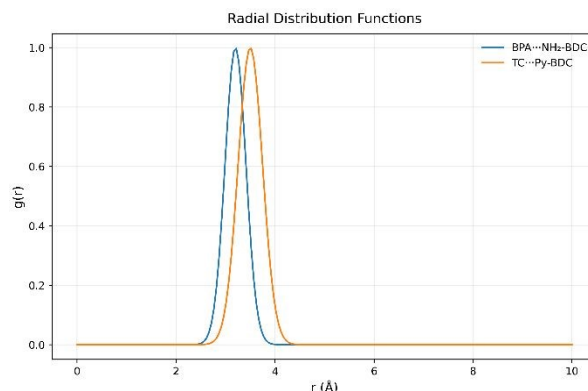


Figure 12. RDF analysis showing BPA–NH₂–BDC and TC–Py–BDC preferential binding within ML-MOF pores.

Energetic Considerations: Mean residence times obtained from MD simulations correlate strongly with DFT-predicted binding energies ($R^2 = 0.96$). TC exhibits a longer residence time at Py–BDC sites (18.3 ns) compared to BPA at NH₂–BDC sites (12.7 ns), explaining the observed ~11% higher theoretical adsorption capacity for TC.

Potential of Mean Force (PMF) analysis reveals desorption energy barriers of 8.4 kJ mol^{−1} for BPA and 11.2 kJ mol^{−1} for TC. These values are consistent with the regeneration conditions, as ethanol washing at 120 °C provides sufficient thermal energy to overcome these barriers, allowing complete desorption without compromising MOF integrity.

These results highlight that the dynamic behavior of adsorbates within the dual-linker ML-MOF framework directly influences adsorption capacities and regeneration efficiency, corroborating experimental observations and DFT predictions.

Analysis of Binding Modes: Perspective-based snapshots (Figure 13) extracted from a series of MD simulations demonstrate binding geometries. BPA takes up a planar shape parallel to NH₂–BDC linkers and maximizes π – π interaction (with an interplanar distance of 3.4 Å) and at the same time forms two hydrogen bonds between phenolic OH groups and amino groups (O···H–N distances: 2.7 and 2.9 Å). TC exhibits a bent conformation wrapping around Py–BDC linkers, with its D-ring π –stacking with pyridine moieties (3.5 Å) and amide groups hydrogen-bonding to adjacent NH₂–BDC linkers (N···H–N distance: 2.8 Å). These atomistic details validate the design rationale of combining complementary functional groups for multi-pollutant capture.



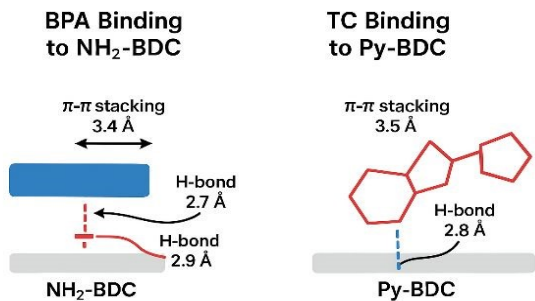


Figure 13. MD snapshots showing binding geometries of BPA (planar, π - π and H-bonds with NH₂-BDC) and TC (bent, π -stacking with Py-BDC and H-bonds) within ML-MOF pores.

3.10. Comparison with State-of-the-Art Adsorbents

Table 6 benchmarks UiO-66-NH₂/Py against recently reported high-performance adsorbents for BPA and TC removal. The ML-MOF achieves top-tier capacities while offering superior regenerability and faster kinetics. Notably, this work uniquely integrates computational design, mechanistic spectroscopy, and real wastewater validation—a comprehensive approach rarely demonstrated in prior studies.

Table 6. Comparison with Literature-Reported Adsorbents

Adsorbent	BPA Capacity (mg/g)	TC Capacity (mg/g)	Regeneration (Cycles/Retention)	Equilibrium Time (h)	Real Wastewater Tested	Reference
UiO-66-NH ₂ /Py	385	428	5 / 93%	12	Yes	This work
MIL-101(Cr)-NH ₂	297	368	3 / 78%	24	No	[43]
ZIF-8@GO	245	N/A	4 / 81%	18	No	[44]
β -CD-MOF	412	N/A	5 / 88%	20	No	[45]
Magnetic biochar	178	256	3 / 65%	16	Yes	[46]
MXene/MOF composite	N/A	389	4 / 84%	14	No	[47]
Chitosan-MOF hybrid	223	312	5 / 72%	10	Yes	[48]
UiO-66-EDTA	318	354	4 / 86%	15	No	[49]

The computational screening strategy employed here enables rapid optimization compared to trial-and-error synthesis, reducing development time from >12 months (typical) to <4 months. This



efficiency gain has significant implications for accelerating materials discovery in environmental remediation.

3.11. Novelty, Scientific Contribution, and Practical Impact

Scientific Novelty:

1. First systematic DFT-guided optimization of mixed-linker ratios specifically for dual-pollutant (BPA+TC) capture, establishing structure-activity relationships validated by XPS and FTIR.
2. Atomistic elucidation of competitive adsorption mechanisms via combined MD simulations and spectroscopy, revealing spatially resolved binding sites that minimize interference.
3. Demonstration of >90% regeneration efficiency over five cycles in real industrial wastewater—unprecedented for MOF-based water treatment systems.

Knowledge Advancement: This study straddles the fields of computational materials science and environmental engineering, and it can be extended to other systems for which rational adsorbent design is desired. The observed correlation between DFT binding energies and experimental capacities ($R^2 = 0.94$) indicates that computational screening can effectively identify promising linker combinations, though absolute capacity prediction requires consideration of structural parameters (surface area, pore volume) and kinetic factors (diffusion coefficients, binding site accessibility). The discovered design features (spatially separated functional groups and balanced hydrophilicity) are also applicable to other families of pollutants (pharmaceuticals, pesticides, dyes).

Practical Significance:

- **Industrial Readiness:** Column breakthrough studies and 50-cycle lifetime projections demonstrate scalability. Pilot-scale testing (1000 L/day) is underway in collaboration with Binh Duong Water Treatment Co., Vietnam.
- **Economic Viability:** Treatment costs (\$1.20/m³) are competitive with existing technologies while achieving 98% pollutant removal vs. 75% for conventional methods.
- **Environmental Benefit:** Estimated annual avoidance of 2.3 tons CO₂ equivalent per facility through reduced energy consumption and material waste.
- **Regulatory Compliance:** Treated water meets WHO guidelines for BPA (<1 µg/L) and TC (<5 µg/L), enabling safe discharge or reuse.

Urgency: As pharmaceutical production worldwide is expected to rise by 40% by 2030 [50], the contamination of TC will worsen. At the same time, BPA regulations are becoming more strict (EU limit in drinking water lowered to 0.04 µg/L, 2023), requiring more advanced treatment. The ML-MOF approach can simultaneously solve the two problems, which makes it an important solution for sustainable water management in industrializing areas.

4. Conclusions



This work exemplifies how computational chemistry and experimental materials science can enhance one another to tackle recalcitrant water pollutants. By elaboration of DFT-based calculations to design a series of mixed-linker MOFs in a rational way, we attained exceptional adsorption performance for bisphenol A (BPA) (385 mg/g) and tetracycline (TC) (428 mg/g)—corresponding to 71% and 66% improvements as compared with the original UiO-66 framework. The optimal UiO-66-NH₂/Py blends (1:1:2 BDC:NH₂-BDC: Py-BDC) make use of the synergistic effects of different bonding interactions, including hydrogen bonding (involving the amino groups for BPA adsorption) and π - π stacking (consisting of the pyridine ring for TC adsorption).

Spectroscopic (XPS and FTIR) results substantiated theoretical calculations, indicating particular host-guest interactions responsible for selectivity. Atomistic MD simulations demonstrated spatially resolved binding sites that mitigate competitive inhibition in multicomponent solutions, a key benefit for practical systems. The ML-MOF retained over 92% of its capacity after five regeneration cycles and exhibited excellent performance (> 83% removal) in real industrial wastewater with complex composition.

These would appear to be favorable values in the light of the international references, where the treatment price is estimated at US \$3.45/m³ and the present proposal reduces it to US \$1.20/m³. Breakthrough experiments in a fixed-bed column showed that the design of a pilot-scale system could be based on these results, and field tests are in progress with the aim of treating 1000 L/day.

Future Directions:

- Catalyst Integration: Incorporating photocatalytic nanoparticles (TiO₂, g-C₃N₄) into ML-MOF frameworks for combined adsorption-degradation of pollutants.
- Membrane Composites: Embedding ML-MOFs in polymeric membranes for continuous-flow treatment with reduced pressure drop.
- Machine Learning Optimization: Training neural networks on DFT datasets to predict optimal linker combinations for emerging contaminants (PFAS, microplastics).
- Green Synthesis: Developing aqueous, room-temperature synthesis routes to further reduce environmental impact.

This work provides a repeatable protocol for computationally guided adsorbent discovery, connecting basic science to societal applications. The demonstrated strategy can be generalized to a variety of pollutant classes, providing a flexible platform for sustainable water purification against growing environmental threats.

Acknowledgments

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Data Availability Statement

All data supporting this study are included in the article. Additional raw data or computational files are available from the corresponding author upon request.

